



Himilan® SP3000

Purge compound resin

For Extrusion Coating

Description

Himilan® SP3000 is a purge compound resin designed to use after processing of Ionomer and/or acid copolymer in conventional extrusion and co-extrusion coating equipment.

Features/Benefits

- Excellent purge efficiency to purge out Ionomer and/or acid copolymer, especially at the dead space in adapter and/or die
- Easily switched from SP3000 to PE
- Shorten purge time compared to PE purge
- Much less fume and odor compared to conventional purge compounds

Table 1
Properties of Himilan® SP3000

Resin Property	Typical Value	Test Method
Density, kg/m ³	1,030	ISO 1183
Melt Flow Rate, g/10min	2	ISO 1133 Cond. 190°C/2.16 kg
Melting Point(DSC), °C	110	ISO 3146
Freezing Point(DSC), °C	99	ISO 3146

Caution: These values are not standard specification.

Processing Information

A typical procedure for use of *Himilan®* SP3000 is shown in Figure 1. Actual purge procedure will usually be determined by either the specific equipment or the type of Ionomer and/or acid copolymer.

Figure 1. Typical procedure (Ionomer/Acid copolymer to LDPE) for 90mm Φ Extruder

Process	Resin	Setting Temperature	Screw speed	Notes
Production (Ionomer / Acid copolymer)	Ionomer / Acid copolymer	C1=180°C C2=260°C C3→D=280-300°C	100rpm	
Discharge residual pellets				Discharge all pellets in a hopper
Charge purge resin	SP3000 [50kg]	C1=200°C C2=260°C C3→D=280-300°C	100rpm	Keep the temperature (except C1) until SP3000 comes out across the die (full width). Often move deckle in and out. Pay attention to motor load and resin pressure.
Temperature rising		C1=200°C C2=280°C C3→D=320-340°C	10rpm	When SP3000 comes out across the die (full width), reduce screw speed minimum (e.g. 10rpm) and raise temperature.
Purging			200rpm	Raise screw speed after temperature is reached at the designed profile. Often move deckle in and out. Pay attention to motor load and resin pressure.
Charge LDPE	LDPE [50kg]	C1=200°C C2=280°C C3→D=320-340°C	200rpm	Often move deckle in and out.
Confirmation of purge completion			100rpm	Lower screw speed, move deckle out and fix it. Then, confirm the web appearance if purge has been completed.
Nest Production	Next Resin (e.g.: LDPE)	As appropriate	100rpm	

In order to improve purge efficiency;

- Increase screw speed as much as possible.
- Often move deckle in and out, however, close to the end of purge, move deckle out and fix it.
- Occasionally change screw speed up and down.
- Use appropriate purge resin under the right conditions.

Safety

SP3000 as supplied by Dow is not considered hazardous materials. As with any hot material, care should be taken to protect the hands and other exposed parts of the body when handling molten polymer. At the recommended processing temperature, certain amounts of fumes evolve from the resin. When resins are overheated, more fumes may occur. Adequate ventilation should be provided to remove fumes from the work area

During the purging, sometimes strong foaming may occur in the melt web, keep away from the die and pay attention to foaming.

For more detailed information on the safe handling Material Safety Data Sheet can be obtained from Dow serving you.